

BACKLOG BUDDY ? STUDY MATERIAL

Reinforcement Learning ? Unit-wise Lecture Notes

Subject Code: AI501 Semester: 6

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Condensed notes covering all five units: Markov Decision Processes, Dynamic Programming for RL, Temporal Difference Learning, Policy Gradient Methods, Deep Reinforcement Learning.

Unit 1: Markov Decision Processes

- Definition, key concepts and significance of Markov Decision Processes
- Important diagrams / derivations / algorithms for Markov Decision Processes
- Solved examples and previous-year questions on Markov Decision Processes

Unit 2: Dynamic Programming for RL

- Definition, key concepts and significance of Dynamic Programming for RL
- Important diagrams / derivations / algorithms for Dynamic Programming for RL
- Solved examples and previous-year questions on Dynamic Programming for RL

Unit 3: Temporal Difference Learning

- Definition, key concepts and significance of Temporal Difference Learning
- Important diagrams / derivations / algorithms for Temporal Difference Learning
- Solved examples and previous-year questions on Temporal Difference Learning

Unit 4: Policy Gradient Methods

- Definition, key concepts and significance of Policy Gradient Methods
- Important diagrams / derivations / algorithms for Policy Gradient Methods
- Solved examples and previous-year questions on Policy Gradient Methods

Unit 5: Deep Reinforcement Learning

- Definition, key concepts and significance of Deep Reinforcement Learning
- Important diagrams / derivations / algorithms for Deep Reinforcement Learning
- Solved examples and previous-year questions on Deep Reinforcement Learning

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.